

Final Project Abstract: Credit Card Fraud Detection Using Data Mining Techniques

CPT_S-315: Introduction to Data Mining

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Introduction and Motivation

Credit card fraud remains a major issue in modern financial systems, resulting in billions of dollars in losses each year. Detecting fraudulent transactions is especially difficult because they are rare and constantly changing as fraudsters adapt their methods. In this project, we use data mining and machine learning techniques to identify fraudulent transactions within a real-world dataset. Using the Kaggle Credit Card Fraud dataset, which contains over 280,000 transactions, we aim to build a model that can accurately distinguish between legitimate and fraudulent activity.

Dataset Description

The dataset includes anonymized features (V1–V28), along with transaction time, transaction amount, and a label indicating whether a transaction is fraudulent. Because the data has been anonymized, we cannot directly interpret what each feature represents. Another major challenge is that fraudulent transactions make up less than 1% of the dataset, creating a strong imbalance between normal and fraudulent cases. This makes simple accuracy less useful, so we instead focus on measures like precision, recall, and F1-score to better evaluate performance.

Methodology

Our approach begins with data preprocessing, including normalizing transaction amounts and exploring the data to understand patterns and relationships. We will look for trends and correlations that may help distinguish fraudulent behavior. To handle the imbalance in the dataset, we will apply data balancing techniques and class weighting to prevent the model from favoring non-fraudulent predictions.

We plan to test and compare several classification models, including logistic regression, decision trees, and random forests. These models will be evaluated based on how well they detect fraud while minimizing false positives. In addition, we may explore anomaly detection by treating fraudulent transactions as unusual patterns compared to normal behavior.

Challenges

One of the main challenges in this project is that the dataset uses anonymized features, which makes it difficult to understand what each variable represents in real-world terms. This limits our ability to clearly explain why a transaction is classified as fraudulent. Another challenge is finding the right balance between detecting fraud and avoiding false alarms, since incorrectly flagging normal transactions can inconvenience users.

Expected Outcomes

The goal of this project is to compare different data mining approaches for fraud detection and identify which methods perform best on highly imbalanced data. We expect to develop a model that can effectively detect fraudulent transactions while maintaining reasonable accuracy on normal transactions. This project demonstrates how data mining techniques can be applied to real-world financial problems, particularly those involving rare but important events.

Works Cited

[1] ULB Machine Learning Group. Credit Card Fraud Detection Dataset. Kaggle.

<https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud/data>